

When Does a Machine Learning Model Fail? Linking Feature-Space Changes to Prediction Degradation

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Abstract—Deployed models are routinely monitored with drift detectors that raise an alarm whenever the input distribution changes, yet a change in $P(X)$ need not change the error of the model. We ask which measurable feature-space changes actually predict prediction degradation, and why the rest do not. We assemble a testbed of 173 natural source→target shifts drawn from seven public tabular datasets (demographic, geographic, temporal and carrier splits), train three model classes with three seeds each (1,557 evaluations), and compare thirteen label-free signals against the true accuracy drop. We also propose *Importance-Weighted Shift* (IWS), which weights each feature’s drift by the model’s own reliance on that feature, and a reweighting decomposition that separates the covariate part of a drop from the conditional part. Three findings stand out. First, drift is nearly universal but weakly informative: a domain classifier separates source from target with $\text{AUC} \geq 0.95$ in 68% of pairs, while 25% of drifted evaluations show no harm and 48% of low-drift evaluations still fail. Second, IWS gives the best harm detection among single signals (AUROC 0.70, AUPRC 0.86) and the highest rank correlation with the drop ($\rho = 0.42$), whereas confidence-based estimators that work well on images correlate weakly on tabular shifts ($\rho \approx 0.07$). Third, in 76% of harmful evaluations the conditional part $P(Y|X)$ dominates the covariate part, and controlled experiments confirm that such concept shift is invisible to every label-free signal. We release code and results to support reproducible monitoring research.

Index Terms—distribution shift, concept drift, model monitoring, performance estimation, tabular data, feature importance

I. INTRODUCTION

A model that was accurate at validation time can quietly lose accuracy after deployment because the data it receives no longer resembles the data it was trained on [1], [2]. Labels that would reveal the loss usually arrive late or never, so practitioners rely on *drift detectors*: univariate tests such as Kolmogorov–Smirnov (KS) or the Population Stability Index (PSI), kernel two-sample tests [3], or classifier two-sample tests [4], [5]. These tools answer the question “has $P(X)$ changed?”. The question an operator actually needs answered is different: “will my model now make more mistakes?”.

The two questions come apart for two reasons. A change confined to features the model ignores is harmless, yet it

triggers every detector. Conversely, a change in the relation between features and label, $P(Y|X)$, can destroy accuracy while leaving $P(X)$ untouched, so no detector fires. Recent work on tabular benchmarks found that such conditional shifts are common in practice [6], and that the accuracy gap tracks label-distribution shift [7]. What is missing is a direct, large-scale measurement of how well each label-free signal predicts the actual drop, and of *why* it fails when it does.

This paper provides that measurement and makes four contributions:

- 1) **A shift–degradation testbed.** 173 natural source→target pairs from seven public tabular datasets, three model classes, three seeds, 1,557 evaluations, each with its true accuracy drop and 13 label-free signals.
- 2) **Importance-Weighted Shift (IWS).** A simple model-aware statistic that weights per-feature drift by permutation importance. It is the best single predictor of harm in our testbed and costs one importance computation per deployed model.
- 3) **A covariate/conditional decomposition** of each drop by density-ratio reweighting, validated on synthetic shifts with known ground truth, showing that conditional shift dominates natural tabular failures.
- 4) **A failure taxonomy** (stable, benign drift, harmful covariate shift, conditional failure) with practical guidance on when a drift alarm should trigger retraining.

II. RELATED WORK

Shift detection. Classical concept-drift research monitors streams for changes [8]–[11]. For batch shift, Rabanser *et al.* [5] benchmarked two-sample tests on learned representations and found domain classifiers and MMD [3] effective at *detecting* shift. Detection, however, is not harm: they report shift presence, not its effect on error.

Label-free performance estimation. Several methods predict target accuracy from unlabeled data: Average Thresholded Confidence (ATC) [12], difference of confidences [13], agreement-on-the-line [14], [15], and importance-weighted estimators under covariate shift [16], [17], recently extended

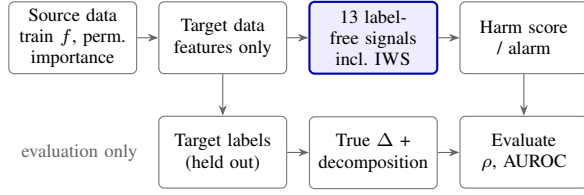


Fig. 1. Study pipeline. The top row is what a deployed monitor sees; the bottom row exists only to score how well each signal predicts the real drop.

with calibration in PAPE [18]. Most were validated on image benchmarks such as WILDS [19]; their behaviour on natural tabular shifts with strong conditional components is less understood.

Harmful-shift testing. Podkopaev and Ramdas [20] test for a rise in risk with labels; Detectron [21] and Amoukou *et al.* [22] detect harmful shift without labels. These methods output an alarm, not an explanation of which features or which kind of shift caused it.

Explaining shifts and drops. Zhang *et al.* [23] attribute a drop to shifted causal mechanisms with Shapley values; DISDE [24] splits a drop into harder examples, conditional shift and unseen examples; Kulinski and Inouye [25] explain shifts via transport maps. Explanation shift [26] monitors SHAP [27] distributions, while FMMO [28] shows local attributions can stay stable as error rises. Our work is complementary: rather than proposing one more detector, we measure which signals track degradation across many natural shifts, and we explain the misses with a decomposition.

III. PROBLEM FORMULATION

Let $P_S(X, Y)$ be the source distribution on which a classifier f is trained, and $P_T(X, Y)$ a deployment distribution. We observe labelled source validation data $\{(x_i, y_i)\}_{i=1}^{n_S}$ and unlabelled target features $\{x_j\}_{j=1}^{n_T}$. The **accuracy drop** is

$$\Delta = \text{Acc}_S(f) - \text{Acc}_T(f), \quad (1)$$

and a shift is **harmful** if $\Delta > \tau$ (default $\tau = 2$ pp; Sec. VI-G varies τ). Writing $P(X, Y) = P(X)P(Y|X)$, a *covariate* shift changes only $P(X)$, a *conditional* (concept) shift changes $P(Y|X)$, and a *label* shift changes $P(Y)$ with $P(X|Y)$ fixed [2], [29]. A **signal** is any statistic $s(f, \mathcal{D}_S, \{x_j\})$ computed without target labels. Our goal is to measure how well each s ranks pairs by Δ and separates harmful from benign shifts.

IV. METHOD

Fig. 1 summarizes the study: label-free signals are computed exactly as a monitor would compute them; target labels are used only afterwards, to score the signals.

A. Label-free signals

We group thirteen signals by what they look at.

Data-only (D). Mean and maximum per-feature KS statistic; mean per-feature Wasserstein-1 distance on source-standardized features; mean PSI with ten source-quantile bins;

unbiased MMD² with an RBF kernel and median-heuristic bandwidth [3]; and the AUC of a domain classifier (gradient-boosted trees, 3-fold cross-fitting) that separates source from target rows [4], [5].

Model-aware drift (M), proposed. Let d_j be the drift of feature j (KS statistic or standardized Wasserstein distance) and I_j the permutation importance [30], [31] of feature j for model f , measured as the accuracy loss on source validation data when feature j is shuffled. With $w_j = \max(I_j, 0) / \sum_k \max(I_k, 0)$,

$$\text{IWS}(f) = \sum_{j=1}^p w_j d_j. \quad (2)$$

IWS is zero when only ignored features move and large when the features the model relies on move. It needs source labels once (for w) and no target labels. We report IWS-KS and IWS-Wass. The following observation motivates the weighting.

Proposition 1. Let $U \subseteq \{1, \dots, p\}$ be the features f depends on, so $f(x) = g(x_U)$, and suppose $P_T(Y|X) = P_S(Y|X)$ with $P(Y|X)$ depending on x only through x_U . If the shift leaves the marginal of X_U unchanged, $P_T(x_U) = P_S(x_U)$, then $\Delta = 0$, however large the drift of the features outside U .

Proof. $\text{Acc}_D(f) = \mathbb{E}_{x_U \sim P_D} [P(Y=g(x_U) | x_U)]$ for $D \in \{S, T\}$. The integrand is the same function under S and T , and so is the distribution it is integrated against; hence $\text{Acc}_S = \text{Acc}_T$. $\square$

Data-only detectors respond to the drift of all p features and so cannot respect this invariance; IWS approximates it by down-weighting features with zero importance. The converse does not hold: drift in X_U can lower, leave unchanged or even raise accuracy (Sec. VI-C), and a change in $P(Y|X)$ is invisible to any function of X alone.

Model-output (O). Drop in mean maximum class probability, i.e. difference of confidences (DoC) [13]; absolute change in predicted positive rate; ATC drop [12], i.e. source accuracy minus ATC's target estimate; importance-weighted drop (IW-drop), i.e. source accuracy minus the density-ratio-weighted source accuracy [16], [17]; and the increase in disagreement between f and a reference model of a different class [14].

B. Covariate/conditional decomposition

The domain classifier gives out-of-fold probabilities $\hat{p}(T|x)$ for source rows, from which we form density-ratio weights $\hat{r}(x) = \frac{\hat{p}(T|x)}{1-\hat{p}(T|x)} \cdot \frac{n_S}{n_T}$ (probabilities clipped to $[0.01, 0.99]$, weights capped at 50). The weighted source accuracy

$$\widehat{\text{Acc}}_{S \rightarrow T} = \frac{\sum_i \hat{r}(x_i) \mathbb{I}[f(x_i) = y_i]}{\sum_i \hat{r}(x_i)} \quad (3)$$

estimates what the target accuracy would be if only $P(X)$ had changed. We split

$$\Delta = \underbrace{\text{Acc}_S - \widehat{\text{Acc}}_{S \rightarrow T}}_{\text{covariate part}} + \underbrace{\widehat{\text{Acc}}_{S \rightarrow T} - \text{Acc}_T}_{\text{residual (conditional) part}}. \quad (4)$$

Algorithm 1 Computing IWS for a deployed model**Require:** model f , source validation (X_S, y_S) , target batch X_T

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1:  $a \leftarrow \text{Acc}(f, X_S, y_S)$ 
2: for  $j = 1, \dots, p$  do ▷ once per model
3:    $I_j \leftarrow a - \text{Acc}(f, \text{shuffle}_j(X_S), y_S)$ 
4: end for
5:  $w_j \leftarrow \max(I_j, 0) / \sum_k \max(I_k, 0)$ 
6: for each new target batch  $X_T$  do
7:    $d_j \leftarrow \text{KS}(X_S[:, j], X_T[:, j])$  for all  $j$ 
8:   return  $\text{IWS} = \sum_j w_j d_j$ 
9: end for

```

TABLE I
DATASETS AND NATURAL SHIFT AXES.

Dataset	Rows	Feat.	Shift axes (domains)	Pairs
Adult [32]	45,222	48	sex, race, age, education, workclass	34
COMPAS [33]	6,172	13	race, sex, age group	14
Housing [34]	20,428	15	ocean proximity, north/south	14
Bank [35]	41,188	62	time (6 blocks, 2008–10)	15
Elec2 [36]	45,312	6	time (8 blocks)	28
Airlines [37]	381,161	5	carrier (top 8)	56
Covertypes [38]	581,012	54	wilderness area (4)	12

The covariate part is label-free (it equals IW-drop); the residual needs target labels and is used only for analysis. When source and target barely overlap, reweighting cannot represent target regions absent from the source, so the residual also absorbs out-of-support error, which DISDE [24] treats as a separate term. We therefore also report results on the subset with domain-classifier AUC < 0.95 .

C. Harm score

To test whether signals combine, we fit a gradient-boosted regressor that maps all thirteen signals to Δ . It is evaluated *leave-one-dataset-out*: the regressor never sees any pair from the dataset it is scored on.

V. EXPERIMENTAL SETUP

A. Datasets and natural shifts

We use seven public tabular datasets with natural domain structure (Table I). Each split attribute defines domains; every ordered pair of domains with at least 400 rows becomes a source→target pair, except temporal splits, where only past→future pairs are kept. The split attribute itself is removed from the features. Bank Marketing has the known leakage variable `duration` removed and its calendar month removed for the time split. This yields 173 pairs.

B. Models and protocol

We train logistic regression, a histogram gradient-boosted tree ensemble (GBDT, the strongest model family on tabular data [39], [40]) and a two-layer MLP (128–64 units) with scikit-learn [41]. Per source domain, up to 5,000 rows are held out for validation and up to 20,000 used for training; up to 5,000 target rows are sampled. Every pair is run with three

seeds that change the sampling and model initialization, giving $173 \times 3 \times 3 = 1,557$ evaluations.

C. Synthetic suite

To validate signals and the decomposition against known ground truth, we draw $X \sim \mathcal{N}(0, I_{10})$ and $P(Y=1|x) = \sigma(2[1.5x_1 + 2\sin(1.5x_2) + x_1x_3 - 0.3])$, so only x_1, x_2, x_3 matter. Five shift families are applied at magnitudes $m \in \{0, 0.25, 0.5, 1, 1.5, 2\}$: (a) mean shift $+m$ of the seven ignored features; (b) mean shift $+m$ of the three used features; (c) spread: x_2, x_3 scaled by $1+m$, forcing extrapolation; (d) concept shift, where the x_1 coefficient is reduced by $1.5m$ with $P(X)$ fixed; and (e) label shift, where $P(Y=1)$ is moved to $0.5+0.2m$ by class-conditional resampling.

D. Metrics

Spearman ρ between signal and Δ , pooled over all evaluations (95% bootstrap CI, 1,000 resamples) and averaged within each dataset×model×seed group; AUROC and AUPRC for detecting harmful shifts; and the false-positive rate at 90% recall of harmful shifts (FPR@90). Paired comparisons use Wilcoxon signed-rank tests over the 63 groups with Holm correction.

VI. RESULTS

A. How common is harmful drift?

Shift is the norm, not the exception. Across the 1,557 evaluations the mean accuracy drop is 10.9pp; 71% exceed $\tau = 2$ pp, while 17% show a *gain* in accuracy. A domain classifier separates source from target with AUC ≥ 0.75 in 87% of evaluations and ≥ 0.95 in 68%. A detector thresholded at AUC 0.75 therefore alarms almost always, yet 25% of those alarms are benign (339 of 1,353). Among the 204 evaluations with little drift, 48% are still harmful: silent failures that no data-only detector can see. The alarm raises the harm rate only from 48% to 75%.

B. Which signals track degradation? (RQ1)

Table II ranks all signals. The model-aware IWS variants lead: IWS-Wass has the highest pooled and within-group correlation ($\rho = 0.42$ and 0.36), and IWS-KS the best harm detection (AUROC 0.70, AUPRC 0.86). Weighting by importance roughly doubles the correlation of plain KS ($0.22 \rightarrow 0.40$) and raises that of mean Wasserstein from 0.31 to 0.42. Fig. 2 shows why the domain classifier, despite a similar pooled ρ , is a poor alarm: most pairs pile up at AUC ≈ 1 , where it cannot rank them. Within that saturated subset IWS-KS still ranks drops better ($\rho = 0.30$ vs. 0.22), and on unsaturated pairs the gap widens (0.30 vs. 0.15).

Confidence-based estimators, which are among the best label-free accuracy predictors on image benchmarks [12], [13], correlate weakly here: ATC drop $\rho = 0.07$ and DoC $\rho = 0.07$. Paired tests over the 63 groups confirm that IWS-KS beats PSI and MMD after Holm correction ($p_{\text{Holm}} < 10^{-4}$ and 0.022). Its advantages over mean KS (38 of 63 groups; pooled $\Delta\rho = +0.18$, 95% CI $[0.13, 0.23]$) and over ATC (pooled

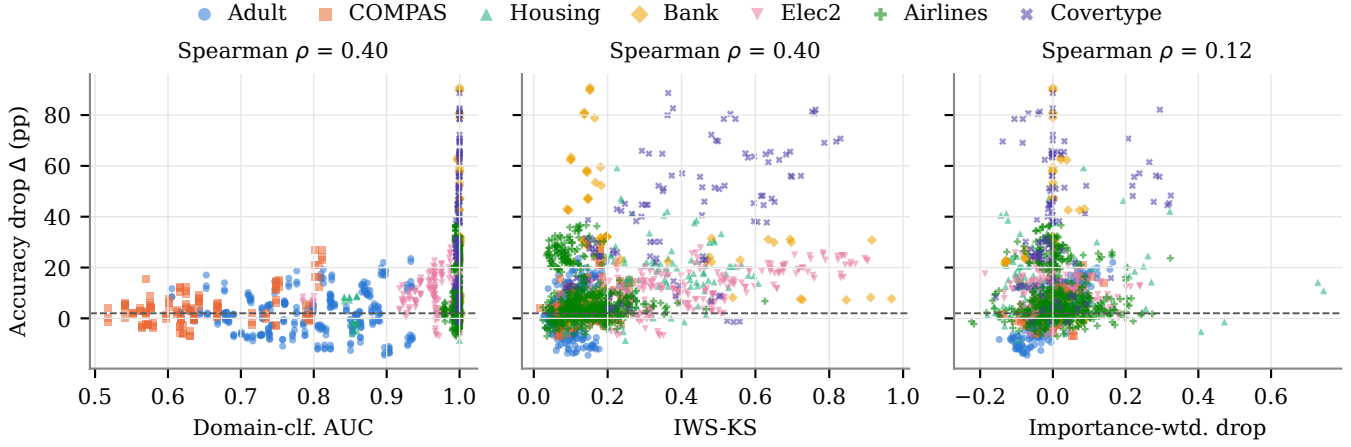


Fig. 2. Accuracy drop versus three signals for every evaluation. The domain-classifier AUC saturates at 1 for most natural shifts, so it cannot rank them; IWS-KS spreads the same pairs out. The density-ratio estimate (right) is close to zero for most harmful pairs because their loss comes from $P(Y|X)$. Dashed line: harm threshold $\tau = 2$ pp.

TABLE II

RANK CORRELATION OF EACH LABEL-FREE SIGNAL WITH THE ACCURACY DROP AND DETECTION OF HARMFUL SHIFTS ($\Delta > 2$ PP), 1,557 EVALUATIONS. D = DATA-ONLY, M = MODEL-AWARE, O = MODEL-OUTPUT; $\dagger$ PROPOSED. BEST IN BOLD.

Signal	ρ pooled [95% CI]	$\bar{\rho}$ within	AUROC	AUPRC
IWS-Wass. [†] (M)	0.42 [0.37, 0.46]	0.36	0.69	0.84
Domain-clf. AUC (D)	0.40 [0.35, 0.44]	0.29	0.66	0.81
IWS-KS [†] (M)	0.40 [0.35, 0.44]	0.34	0.70	0.86
KS (max) (D)	0.35 [0.30, 0.39]	0.28	0.62	0.80
MMD ² (D)	0.33 [0.28, 0.37]	0.26	0.63	0.82
Wasserstein (mean) (D)	0.31 [0.26, 0.34]	0.27	0.65	0.82
Prediction shift (O)	0.28 [0.23, 0.33]	0.27	0.60	0.81
Disagreement incr. (O)	0.25 [0.20, 0.30]	0.29	0.63	0.82
PSI (mean) (D)	0.23 [0.17, 0.27]	0.22	0.65	0.80
KS (mean) (D)	0.22 [0.18, 0.26]	0.30	0.64	0.81
Importance-wtd. drop (O)	0.12 [0.07, 0.17]	0.18	0.62	0.79
ATC drop (O)	0.07 [0.02, 0.12]	0.08	0.58	0.80
Confidence drop (DoC) (O)	0.07 [0.02, 0.12]	0.07	0.60	0.82

$\Delta\rho = +0.33$, CI [0.26, 0.39]) have pooled bootstrap intervals that exclude zero but do not survive Holm correction at the group level ($p_{\text{Holm}} = 0.11$ and 0.10). Against the domain classifier, the pooled correlations are equal ($\Delta\rho = 0.00$), and the AUROC/AUPRC advantage is not significant. We therefore claim that IWS is the most reliable single signal, not that it dominates every alternative.

No signal wins everywhere. Table III breaks correlations down by dataset. Confidence-based signals are excellent on Adult (ATC $\rho = 0.87$) and wrong-signed on Elec2 (-0.50), where the relation between demand and price direction drifts over time and the models stay confident while they fail. IWS-KS is best on Housing and Elec2 and positive on six of seven datasets; Airlines, where carrier changes alter delay patterns rather than inputs, defeats nearly all signals.

TABLE III

WITHIN-DATASET SPEARMAN ρ WITH Δ (ALL MODELS AND SEEDS).

Dataset	KS	DC-AUC	ATC	IW-drop	IWS-KS
Adult	0.02	0.05	0.87	0.81	0.14
COMPAS	0.22	0.32	0.28	0.27	0.26
Housing	0.44	0.38	0.26	0.04	0.53
Bank	0.45	0.31	-0.12	0.01	0.44
Elec2	0.46	0.15	-0.50	-0.01	0.59
Airlines	-0.22	0.20	-0.03	-0.04	-0.13
Covertypes	0.70	0.65	0.01	0.21	0.57

C. Controlled shifts: when does drift hurt? (RQ2)

Fig. 3 reports the synthetic suite (mean over three seeds and three models). (a) Shifting the seven ignored features drives the domain-classifier AUC to 1.0 while the drop stays at 1.0 pp; IWS-KS stays flat at 0.03, so it is the only drift signal that is not fooled. (b) Shifting the used features moves IWS strongly, yet accuracy *improves* by 11.0 pp because the samples move away from the decision boundary. Drift in important features is therefore necessary for covariate-driven harm but not sufficient. (c) When the used features spread out and force extrapolation, the drop reaches 13.6 pp, and IW-drop (10.7 pp) and IWS both track it. (d) Under pure concept shift the drop reaches 28.8 pp while the domain-classifier AUC stays at 0.50: every label-free signal is blind, as theory predicts. (e) Label-prior shift changes $P(X)$ through $P(X|Y)$ but costs at most 1.5 pp.

D. Case study: the same drift statistic, opposite outcomes

Fig. 4 contrasts two natural pairs for GBDT. In Elec2 (time block 0 $\rightarrow$ 7), the Victorian price, Victorian demand and interstate transfer features drift heavily (KS 0.53–0.73), and the mean KS over all features is 0.36. The model, however, relies almost entirely on the New South Wales price ($w = 0.95$, KS only 0.11), and accuracy is unchanged (81.7% $\rightarrow$ 82.4%). In

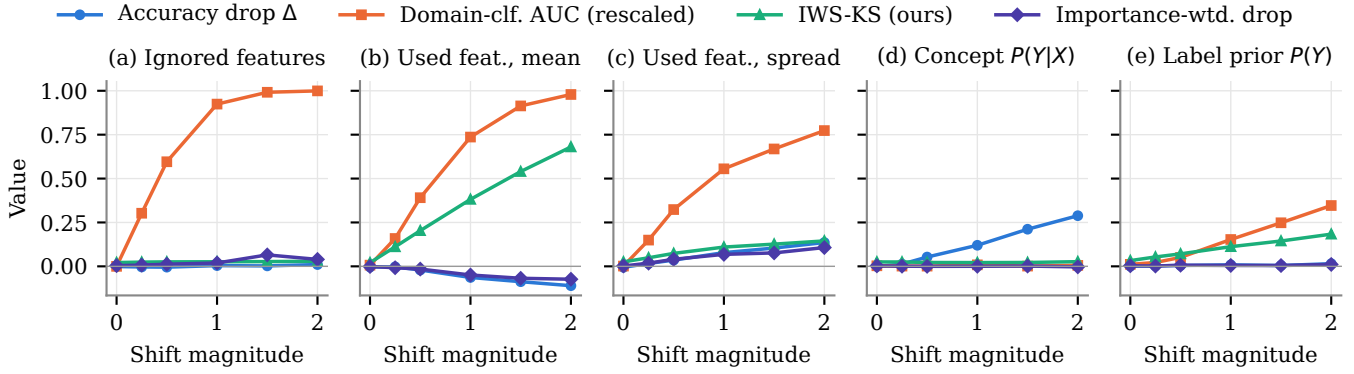


Fig. 3. Synthetic shifts with known ground truth. Data-only drift (orange) reacts to every change in $P(X)$, harmful or not. IWS (green) ignores irrelevant drift (a) but cannot tell helpful from harmful movement of used features (b, c). No label-free signal sees concept shift (d).

TABLE IV

DECOMPOSITION PER DATASET. EVALS = PAIRS $\times$ MODELS $\times$ SEEDS; $\bar{\Delta}_{\text{HARM}}$ = MEAN DROP OF HARMFUL EVALUATIONS (PP); $Y|X$ SHARE = RESIDUAL SHARE OF MEAN ABSOLUTE DROP; $|\pi_T - \pi_S|$ = MEAN CHANGE IN POSITIVE RATE.

Dataset	Evals	Harmful	$\bar{\Delta}_{\text{harm}}$	$Y X$ share	$ \pi_T - \pi_S $
Adult	306	49%	8.8	55%	0.134
COMPAS	126	56%	8.6	69%	0.141
Housing	126	81%	17.4	67%	0.246
Bank	135	79%	26.5	94%	0.125
Elec2	252	86%	13.0	78%	0.043
Airlines	504	72%	11.7	70%	0.117
Covtype	108	97%	41.4	86%	0.267

TABLE V

HARM DETECTION: SINGLE SIGNALS VS. A META-REGRESSOR TRAINED LEAVE-ONE-DATASET-OUT ($\Delta > 2$ PP).

Method	ρ	AUROC	AUPRC	FPR@90
Domain-clf. AUC	0.40	0.66	0.81	0.75
ATC drop	0.07	0.58	0.80	0.95
Importance-wtd. drop	0.12	0.62	0.79	0.83
IWS-KS [†]	0.40	0.70	0.86	0.76
<i>Meta-regressor, leave-one-dataset-out</i>				
Data-only signals	0.13	0.55	0.75	0.84
Data-only + IWS	0.22	0.59	0.78	0.86
All signals (harm score) [†]	0.34	0.67	0.83	0.81

Housing (coastal $\rightarrow$ inland), mean KS is lower (0.20), but the drift falls on latitude, longitude and median income, the three features the model uses most, and accuracy collapses from 88.4% to 61.3%. Mean KS ranks the two pairs the wrong way round (0.36 vs. 0.20), whereas IWS-KS ranks them correctly (0.11 vs. 0.32).

E. What kind of shift causes the failures? (RQ3)

The decomposition behaves correctly where the truth is known: under concept shift it assigns 29.2 ppo of the 28.8 pppdrop to the residual and -0.3 ppto to the covariate part; under spread shift it assigns 10.7 of 13.6 ppto to the covariate part. Its one failure mode is complete separation: with ignored features shifted by $m = 2$ (AUC = 1.0) it attributes a spurious 4.0 ppto covariate shift.

On the natural testbed (Fig. 5, Table IV) the residual part dominates the covariate part in 76% of harmful evaluations and accounts for 55% (Adult) to 94% (Bank) of the mean absolute drop. Restricting to pairs with good overlap (AUC < 0.95 , 271 harmful evaluations) lowers but does not remove the dominance: 62% of these are residual-dominated. This agrees with the finding of Liu *et al.* [6] that $Y|X$ shifts dominate natural tabular shifts, and it explains the weak correlations above: the largest component of most drops is one that label-free signals cannot observe.

F. Can signals be combined into a harm score? (RQ4)

Table V compares single signals with a meta-regressor evaluated leave-one-dataset-out. Learned combinations transfer poorly across datasets: data-only signals reach AUROC 0.55, adding IWS raises this to 0.59, and all thirteen signals reach 0.67 with $\rho = 0.34$, still below IWS-KS alone (0.70, $\rho = 0.40$). Fig. 6 shows IWS-KS holds precision above 0.9 up to about 30% recall, the operating region relevant for a high-confidence “retrain now” alarm. At 90% recall, however, every method raises false alarms on roughly three of four benign shifts ($\text{FPR@90} \geq 0.75$), a direct consequence of the conditional failures documented above.

G. Ablations

Model class. IWS-KS correlates with the drop for every model (GBDT 0.45, MLP 0.42, logistic regression 0.33), whereas ATC works only for GBDT (0.16) and is near zero for the other two (0.02), whose confidence is not informative of their errors under shift. **Harm threshold.** Varying $\tau \in \{1, 2, 5, 10\}$ ppleaves the AUROC of IWS-KS between 0.70 and 0.71, while ATC falls from 0.60 to 0.49 and the domain classifier stays at 0.66–0.69. IWS is thus robust to how “harmful” is defined. **Target sample size.** Monitoring often has to act on small batches. Table VI recomputes four signals for GBDT (seed 0, all 173 pairs) on $n_T \in$

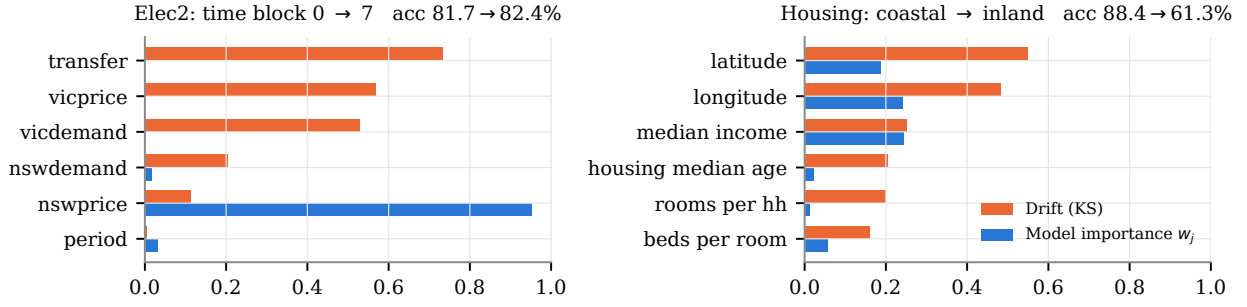


Fig. 4. Per-feature drift (orange) and model importance (blue) for the six most-drifted features of two natural pairs (GBDT). Left: heavy drift in unused features, no harm. Right: moderate drift in the used features, a 27-point accuracy drop.

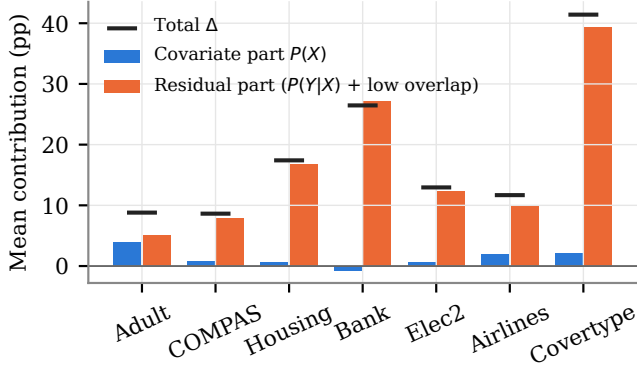


Fig. 5. Mean decomposition of harmful drops (Eq. 4). The residual, conditional part dominates on every dataset.

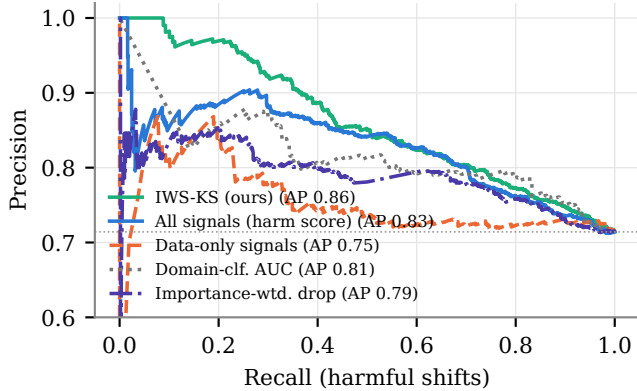


Fig. 6. Precision–recall for detecting harmful shifts. Dotted line: base rate (0.71).

$\{100, 300, 1000, 5000\}$ unlabelled target rows, always scoring against the drop measured on the full sample. IWS-KS keeps the highest rank correlation at every size ($\rho = 0.45$ – 0.47), and its AUROC with only 100 rows (0.73) matches that with 5,000 rows. Mean KS is similarly stable and, in this GBDT-only setting, reaches comparable AUROC (0.72–0.74) at a lower correlation. ATC degrades most at small n_T ($\rho = 0.06$ at 100

TABLE VI
EFFECT OF THE NUMBER OF UNLABELLED TARGET ROWS n_T (GBDT, SEED 0, 173 PAIRS). CELLS: SPEARMAN ρ / AUROC FOR HARM ($\Delta > 2$ PP).

n_T	KS (mean)	DC-AUC	ATC drop	IWS-KS [†]
100	0.28 / 0.72	0.42 / 0.68	0.06 / 0.59	0.47 / 0.73
300	0.30 / 0.74	0.43 / 0.69	0.13 / 0.64	0.46 / 0.72
1,000	0.30 / 0.73	0.43 / 0.69	0.16 / 0.64	0.46 / 0.72
5,000	0.30 / 0.73	0.43 / 0.70	0.15 / 0.64	0.45 / 0.72

TABLE VII
FAILURE REGIMES PER DATASET (SHARE OF EVALUATIONS). HIGH DRIFT = DOMAIN-CLASSIFIER AUC ≥ 0.75 ; HARMFUL = $\Delta > 2$ PP; COVARIATE VS. CONDITIONAL BY THE LARGER TERM OF EQ. (4).

Dataset	Stable	Benign drift	Harmful cov.	Conditional
Adult	19%	33%	21%	28%
COMPAS	39%	6%	10%	45%
Housing	0%	19%	17%	63%
Bank	0%	21%	2%	77%
Elec2	0%	14%	14%	72%
Airlines	0%	28%	24%	48%
Covertypes	0%	3%	10%	87%
All	7%	22%	17%	54%

rows).

VII. DISCUSSION

A taxonomy of failure. Combining drift magnitude (domain-classifier AUC ≥ 0.75), harm ($\Delta > 2$ pp) and the dominant decomposition term places each evaluation in one of four regimes: *stable* (7%), *benign drift* (22%), *harmful covariate shift* (17%) and *conditional failure* (54%). Only the third is what drift detectors are designed for; the second wastes retraining effort and the fourth goes unnoticed. Table VII shows the mix differs sharply by shift axis. Demographic splits (Adult, COMPAS) contain the most stable and benign cases, whereas temporal and geographic splits (Bank, Elec2, Covertypes) are dominated by conditional failures, the regime in which unlabelled monitoring is weakest.

Practical guidance. (i) Replace raw per-feature drift dashboards with importance-weighted drift: it costs one permutation-importance pass per model and filters out drift

in ignored features. (ii) Do not rely on model confidence for tabular monitoring; ATC and DoC were wrong-signed on temporal data. (iii) Because conditional shift dominates, budget for a small stream of delayed labels; no amount of unlabelled monitoring recovers a change in $P(Y|X)$. (iv) Treat a saturated domain classifier as “the data changed”, never as “the model broke”.

Limitations and threats to validity. Our testbed covers seven datasets and binary classification; regression and high-dimensional data may behave differently. Pairs from the same dataset are not independent, which is why we report within-group statistics and leave-one-dataset-out results. The decomposition relies on density-ratio estimation and is unreliable under near-complete separation, as our synthetic check shows. Permutation importance is computed on source data; under strong shift the features the model relies on may change, and conditional importance [27] could refine Eq. (2). Finally, the strongest recent estimators (PAPE [18], Detectron [21]) were not re-implemented; IW-drop and disagreement serve as their simpler counterparts.

VIII. CONCLUSION

Feature-space change predicts model failure only weakly on natural tabular shifts. Drift is nearly universal, a quarter of drift alarms are benign, and most harmful drops come from a change in $P(Y|X)$ that unlabelled data cannot reveal. Weighting drift by what the model actually uses (IWS) is the most reliable label-free signal we tested and is trivial to deploy, but the dominant failure mode calls for monitoring that combines model-aware drift with a small budget of fresh labels. Code, raw results and scripts that regenerate every table and figure are released with the paper.

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